$\lambda = \frac{\lambda_0}{n_1}$ $c = \frac{c_0}{n_1}$ $k = n_1 k_0$ $k = \frac{2\pi}{\lambda}$ $\beta = k_z = k\cos\theta_m$

$\lambda = \frac{c}{\nu}$ $\omega = 2\pi\nu$ $c^2 = \frac{1}{\mu\varepsilon}$ $k_y = k\sin\theta_m = k\frac{m\lambda}{2d} = \frac{m\pi}{d}$

Constantes de propagação

$$\boxed{\beta_m^2 = k^2 - m^2\frac{\pi^2}{d^2}}$$

Condição de auto-consistência

$$\boxed{\sin\theta_m = m\frac{\lambda}{2d}}$$

$$\nabla\times\vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

$\vec{\nabla}\times\vec{E} = -i\mu\omega\vec{H}$

$S_z = \frac{1}{2}\mathrm{Re}(E_x H_y^*)$

$P_z = \int_{-\infty}^{+\infty} S_z\,dy$

$\int_{-\infty}^{+\infty}|U_m(y)|^2\,dy = 1$

$\Rightarrow \int_0^{\frac{d}{2}} U_m(y)^2\,dy + \int_{\frac{d}{2}}^{+\infty} U_m(y)^2\,dy = \frac{1}{2}$

$\int_{-\frac{d}{2}}^{\frac{d}{2}} U_m(y)\,U_\ell(y) = \delta_{m\ell}$

$\frac{e^{i\theta} - e^{-i\theta}}{2i} = \sin\theta$

$\frac{e^{-i\theta} + e^{i\theta}}{2} = \cos\theta$

$\cos(A)\cos(B) = \frac{\sin(A+B) + \sin(A-B)}{2}$

$\sin(A)\sin(B) = \frac{\cos(A-B) - \cos(A+B)}{2}$

$\cos 2\alpha = 1 - 2\sin^2\alpha$

$\sin 2\alpha = 2\sin\alpha\cos\alpha$

## Metais condutores:

Comprimento de onda de corte (acima a guia não suporta modos): $\lambda_c = 2d$

Frequência de corte: $\nu_c = \frac{c}{2d}$ Frequência angular de corte: $\omega_c = \frac{\pi c}{d}$

Guia Monomodal: $1 < M \leq 2$ $1 < \frac{2d}{\lambda} \leq 2 \Leftrightarrow \frac{1}{2} \leq \frac{\lambda}{2d} < 1 \Leftrightarrow d \leq \lambda < 2d$

Nº de modos: $M \doteq \frac{2d}{\lambda}$ $\omega = \omega_c M$

Velocidade de grupo (para modos guiados): $v_m = c\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}} = c\cos\theta_m$

## Dielétricos:

Comprimento de onda de corte: $\lambda_c = 2d\,NA = 2d\sin\bar{\theta}_c$

Frequência de corte: $\nu_c = \frac{1}{NA}\frac{c_0}{2d}$

Frequência angular de corte: $\omega_c = \frac{1}{NA}\frac{\pi c_0}{d}$

Nº de modos: $M \doteq \frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} \Rightarrow M \doteq \frac{2d}{\lambda_0}NA$

Monomodal: $M = 1$ $\frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} < 1$ (Há sempre modos guiados)

Condição de auto-consistência - TE: $\mathrm{tg}\left(\frac{\pi d}{\lambda}\sin\theta - m\frac{\pi}{2}\right) = \sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

Condição de auto-consistência - TM: $\mathrm{tg}\left(\frac{\pi d}{\lambda}\sin\theta - m\frac{\pi}{2}\right) = -\frac{1}{\cos^2\theta_c}\sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

$\sin\theta_c = \frac{n_2}{n_1} \Leftrightarrow \cos\bar{\theta}_c = \frac{n_2}{n_1}$ $\bar{\theta}_c = 90° - \theta_c$

$\sin\theta_a = NA = \sqrt{n_1^2 + n_2^2}$

1. $E_x(y,z) = a_m U_m(y)\,e^{-i\beta_m z}$

Modo TE

$\nabla\times\vec{E} = -i\mu\omega\vec{H} \Leftrightarrow -i\mu\omega\vec{H_y} = -\left(\cancel{\frac{\partial E_z}{\partial x}} - \frac{\partial E_x}{\partial z}\right) \Leftrightarrow \frac{\partial E_x}{\partial z} = -i\mu\omega\vec{H_y} \Rightarrow \frac{\partial(a_m U_m(y)\,e^{-i\beta_m z})}{\partial z} = -i\mu\omega\vec{H_y} \Leftrightarrow \underbrace{a_m U_m(y)\,(-\cancel{i}\beta_m)\,e^{-i\beta_m z}}_{E_x} = -\cancel{i}\mu\omega H_y \Leftrightarrow$

$\Leftrightarrow H_y = \frac{\beta_m}{\mu\omega}E_x \Leftrightarrow H_y = \frac{k\cos\theta_m}{\mu\omega}E_x \Leftrightarrow H_y = \frac{\frac{2\pi}{\lambda}}{\mu\omega}\cos\theta_m E_x \Leftrightarrow H_y = \frac{\frac{2\pi}{\frac{c}{\frac{\omega}{2\pi}}}}{\mu\omega}\cos\theta_m E_x \Leftrightarrow H_y = \frac{\frac{2\pi}{\frac{2\pi c}{\omega}}}{\mu\omega}\cos\theta_m E_x \Leftrightarrow H_y = \frac{\frac{\cancel{\omega}}{c}}{\mu\cancel{\omega}}\cos\theta_m E_x$

$\Leftrightarrow H_y = \frac{\left(\frac{1}{\mu\varepsilon}\right)^{1/2}}{\mu}\cos\theta_m E_x \Leftrightarrow H_y = \frac{1}{\mu\cdot\frac{1}{\mu^{1/2}\varepsilon^{1/2}}}\cos\theta_m E_x \Leftrightarrow H_y = \frac{1}{\frac{\mu^{1/2}}{\varepsilon^{1/2}}}\cos\theta_m E_x$

$S_z = \frac{1}{2}\mathrm{Re}(E_x H_y^*) = \frac{1}{2}\left(a_m U_m(y)\cdot\frac{1}{\sqrt{\frac{\mu}{\varepsilon}}}\cos\theta_m\,a_m U_m(y)\right) = \frac{1}{2}\frac{1}{\sqrt{\frac{\mu}{\varepsilon}}}\cos\theta_m |a_m|^2|U_m(y)|^2$

$P_z = \int_{-\infty}^{+\infty} S_z\,dy = \frac{1}{2}\frac{1}{\sqrt{\frac{\mu}{\varepsilon}}}\cos\theta_m|a_m|^2\overbrace{\int_{-\infty}^{+\infty}|U_m(y)|^2\,dy}^{1} = \frac{|a_m|^2}{2\eta}\cos\theta_m$

2. Onda ascendente: $\Psi_x(y,z) = A_m\,e^{-i(k_{ym}y + \beta_m z)}$

Onda descendente: $\Psi_x'(y,z) = A_m\,e^{-i(-k_{ym}y + \beta_m z)}\,e^{i(m-1)\pi}$

Para TE: $\Psi_x \rightarrow E_x$

$E_x(y,z) = A_m\,e^{-i\beta_m z}\,e^{-ik_{ym}y} + A_m\,e^{-i\beta_m z}\,e^{ik_{ym}y}\,e^{i(m-1)\pi}$

$= A_m\,e^{-i\beta_m z}\left(e^{-ik_y y} + e^{ik_y y}\,e^{i(m-1)\pi}\right)$

$e^{i(m-1)\pi}\begin{cases} 1 & \text{se } m \text{ ímpar} \\ -1 & \text{se } m \text{ par}\end{cases}$

m par:

$E_x(y,z) = A_m\,e^{-i\beta_m z}\left(e^{-ik_y y} - e^{ik_y y}\right)$

$= A_m\,e^{-i\beta z}\left(2i\sin(k_{ym}y)\right)$

$= A_m\left(2i\sin(k\sin\theta_m y)\right)e^{-i\beta_m z}$

$= 2iA_m\left(\sin\left(\frac{m\pi}{d}y\right)\right)e^{-i\beta_m z}$

$k_y = k\sin\theta_m = k\frac{m\lambda}{2d} = \frac{2\pi m\cancel{\lambda}}{2d\cancel{\lambda}} = \frac{m\pi}{d}$

$k = \frac{2\pi}{\lambda}$

$\int_{-\frac{d}{2}}^{\frac{d}{2}} U_m(y)\,U_\ell(y)\,dy = \delta_{m\ell} \Leftrightarrow \int_{-\frac{d}{2}}^{\frac{d}{2}} N^2\sin\left(\frac{m\pi}{d}y\right)\sin\left(\frac{\ell\pi}{d}y\right)dy = \delta_{m\ell} \Leftrightarrow \delta_{m\ell} = \frac{N^2}{2}\left[\int_{-\frac{d}{2}}^{\frac{d}{2}}\cos\left((m-\ell)\frac{\pi}{d}y\right)dy - \int_{-\frac{d}{2}}^{\frac{d}{2}}\cos\left((m+\ell)\frac{\pi}{d}y\right)dy\right]$

$= \frac{N^2 d}{2\pi}\left[\frac{\sin\left((m-\ell)\frac{\pi}{d}y\right)}{(m-\ell)} - \frac{\sin\left((m+\ell)\frac{\pi}{d}y\right)}{(m+\ell)}\right]_{-\frac{d}{2}}^{\frac{d}{2}} = \frac{N^2 d}{\cancel{2}\pi}\left[\frac{\sin\left((m-\ell)\frac{\pi}{2}\right)}{\cancel{(m-\ell)}} + \frac{\sin\left((m-\ell)\frac{\pi}{2}\right)}{(m-\ell)} - \left(\frac{\sin\left((m+\ell)\frac{\pi}{2}\right)}{\cancel{(m+\ell)}} + \frac{\sin\left((m+\ell)\frac{\pi}{2}\right)}{(m+\ell)}\right)\right]$

$= N^2\frac{d}{\pi}\left(\frac{\sin\left((m-\ell)\frac{\pi}{2}\right)}{(m-\ell)} - \frac{\sin\left((m+\ell)\frac{\pi}{2}\right)}{(m+\ell)}\right)$

$\lim_{\ell\to m}\frac{\sin\left((m+\ell)\frac{\pi}{2}\right)}{m+\ell} = \lim_{\ell\to m}\frac{\overbrace{\sin(m\pi)}^{0}}{2m} = 0$

$\lim_{\ell\to m}\frac{\sin\left((m-\ell)\frac{\pi}{2}\right)}{m-\ell}\overset{\frac{0}{0}}{=}\lim_{m-\ell\to 0}\frac{\frac{\pi}{2}\overbrace{\cos\left((m-\ell)\frac{\pi}{2}\right)}^{1}}{1} = \frac{\pi}{2}$

$\frac{N^2 d}{\pi}\left(\frac{\pi}{2} + 0\right) = 1 \Leftrightarrow N^2\frac{d}{2} = 1 \Leftrightarrow N = \sqrt{\frac{2}{d}}$

logo $U_m(y) = \sqrt{\frac{2}{d}}\sin\left(\frac{m\pi}{d}y\right)$

3. a) $E_x(y,z) = A_m\,e^{-i(k_y y + \beta_m z)}$

$E_x\left(\frac{d}{2},z\right) = 0 \Leftrightarrow \underset{\substack{\text{não pode ser 0} \\ \text{senão não havia onda}}}{A_m}\,e^{\overbrace{-i\left(\frac{d}{2}k_y + \beta_m z\right)}^{}} = 0$ Impossível

nunca são 0 porque são imaginários

$-i\frac{d}{2}k_y = \ln(0) \Rightarrow -i\frac{d}{2}k_y \neq -\infty$

b) $E_x\left(\frac{d}{2},z\right) = 0 \Rightarrow A_1\,e^{-ik_{y1}\frac{d}{2}}\,e^{-i\beta_1 z} = -A_2\,e^{-ik_{y2}\frac{d}{2}}\,e^{-i\beta_2 z}$

$A_1 = \mp A_2$, para $\beta_1 = \beta_2$

$k_{y1} = k_{y2}$

4. a) i) $\omega_c = ?$ $\lambda - ?$ $d = 0{,}5\,\mu m$ $n = 1{,}25$

$d \leq \lambda < 2d \Rightarrow 0{,}5\,\mu m \leq \lambda < 1\,\mu m$

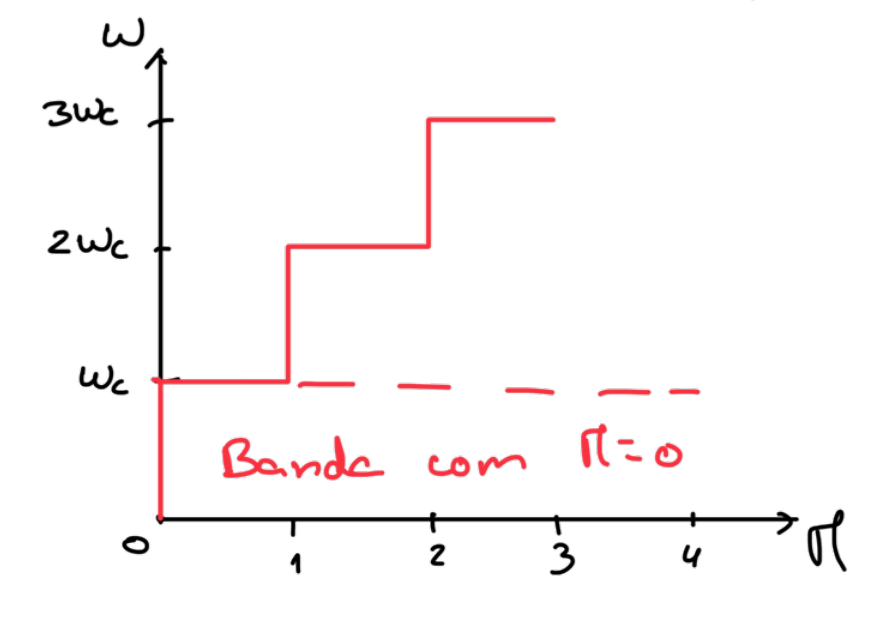
$\omega_c = \frac{\pi c}{d} = \frac{\pi c_0}{n\,d} = \frac{\pi\times 3\times 10^8}{1{,}25\times 0{,}5\times 10^{-6}} = 1{,}51\times 10^{15} = 15{,}1$ PHz

ii) $M \doteq \frac{2d}{\lambda}$ $3 < M \leq 4 \Leftrightarrow 3 < \frac{2d}{\lambda} \leq 4 \Leftrightarrow \frac{1}{4} \leq \frac{\lambda}{2d} < \frac{1}{3} \Leftrightarrow \frac{d}{2} \leq \lambda < \frac{2d}{3} \Leftrightarrow 0{,}25\,\mu \leq \lambda < 0{,}33\,\mu$

$\lambda_c = \frac{2d}{3}$ $\Rightarrow d = \frac{3\lambda_c}{2}$ $\omega_c = \frac{\pi c}{\frac{3\lambda_c}{2}} \Leftrightarrow \omega_c = \frac{2\pi c}{3n\lambda_c} = \frac{2\pi\times 3\times 10^8}{3\times 1{,}25\times 0{,}33\,\mu} = 15{,}2$ PHz

Formulário

b) $\omega = \omega_c M$ $\lambda = 0{,}25\,\mu m$ $M \doteq \frac{2d}{\lambda} = 4 \doteq 3$

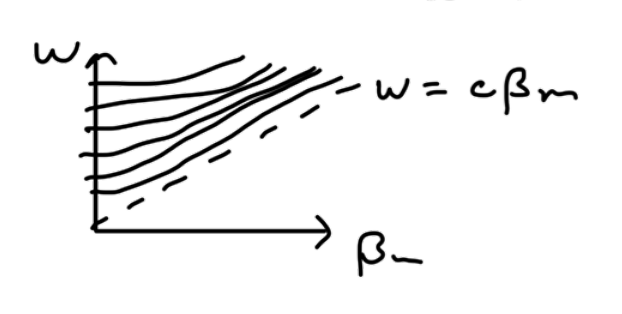


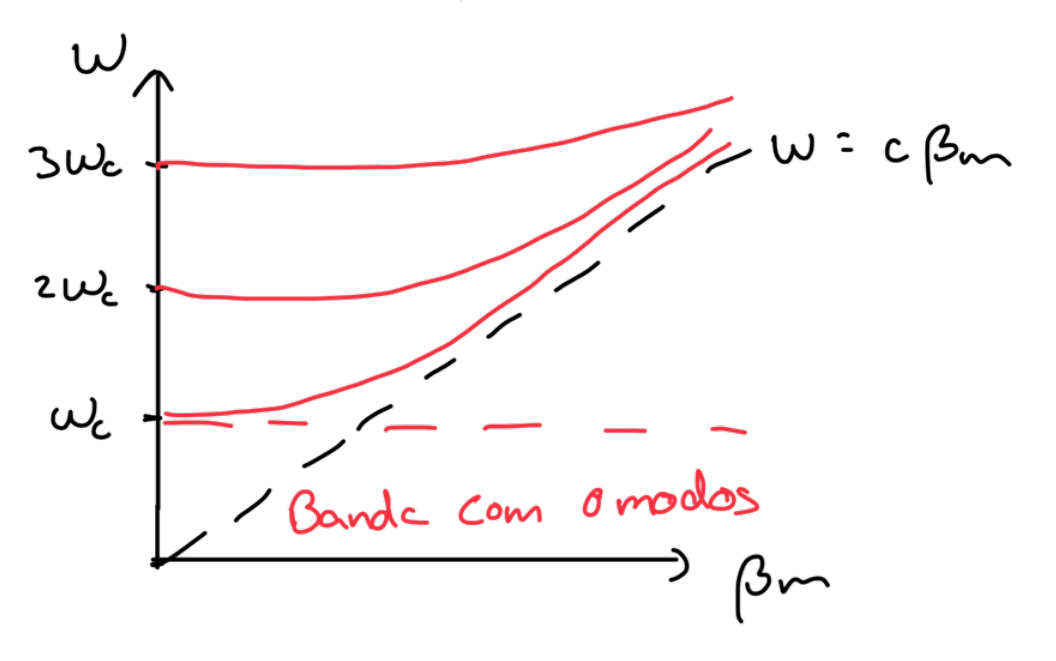
$\omega_c = \frac{\pi c}{d}$

$\beta_m^2 = k^2 - m^2\frac{\pi^2}{d^2} = \left(\frac{2\pi}{\lambda}\right)^2 - \frac{m^2\pi^2}{d^2} = \cancel{(2\pi)^2}\left(\frac{\omega}{\cancel{2\pi}c}\right)^2 - \frac{m^2\pi^2}{d^2} = \frac{\omega^2}{c^2}\left(1 - \frac{m^2\pi^2c^2}{\omega^2d^2}\right) = \frac{\omega^2}{c^2}\left(1 - m^2\frac{\omega_c^2}{\omega^2}\right) \Rightarrow \beta_m = \frac{\omega}{c}\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}}$

$\omega \gg \omega_c \Rightarrow \beta_m = \frac{\omega}{c}\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}} \Rightarrow \beta_m \rightarrow \frac{\omega}{c}$

$\omega \rightarrow m\omega_c \Rightarrow \beta_m = 0$





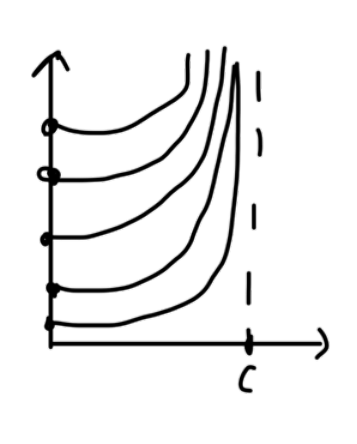
$v = \frac{d\omega}{d\beta} \Leftrightarrow \frac{\partial\beta}{\partial\omega} = \frac{1}{v} \Rightarrow \frac{1}{v} = \frac{\partial}{\partial\omega}\left(\frac{\omega}{c}\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}}\right) = \frac{\partial}{\partial\omega}\left(\frac{1}{c}\sqrt{\omega^2 - m^2\omega_c^2}\right) = \frac{1}{c}\times\frac{1}{2}\left(\omega^2 - m^2\omega_c^2\right)^{-\frac{1}{2}}2\omega$

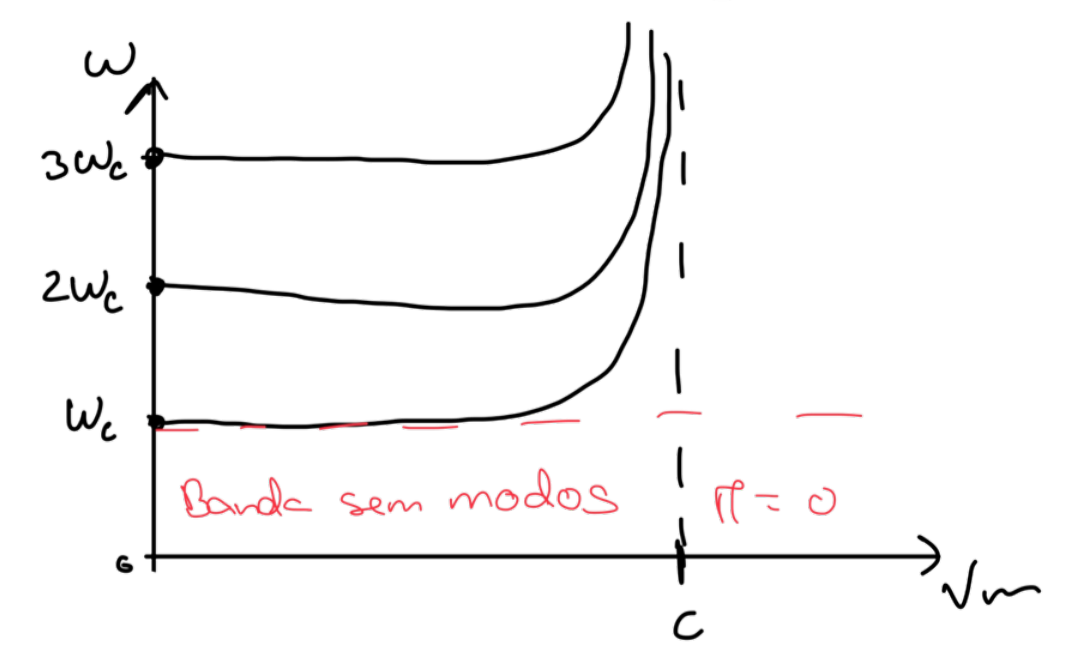
$= \frac{\omega}{c}\left(\omega^2 - m^2\omega_c^2\right)^{-\frac{1}{2}} = \frac{1}{c}\left(1 - m^2\frac{\omega_c^2}{\omega^2}\right)^{-\frac{1}{2}} \Rightarrow v = c\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}}$

$\left(\frac{v_m}{c}\right)^2 = 1 - \frac{m^2\omega_c^2}{\omega^2} \Leftrightarrow 1 - \frac{v_m^2}{c^2} = \frac{m^2\omega_c^2}{\omega^2} \Leftrightarrow \omega^2 = \frac{m^2\omega_c^2}{1 - \frac{v_m^2}{c^2}} \Leftrightarrow \omega = \frac{m\omega_c}{\sqrt{1 - \frac{v_m^2}{c^2}}}$

$\omega \gg \omega_c \Rightarrow v_m = c\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}} \Rightarrow v_m \longrightarrow c$

$\omega \rightarrow m\omega_c \Rightarrow v_m = 0$





5. a) $M \doteq \frac{2d}{\lambda} = 31{,}6 \doteq 31$

b) $\sin\theta_m = \frac{m\lambda}{2d} = m\frac{\frac{\cancel{2\pi c}}{\omega}}{\frac{\cancel{2\pi c}}{\omega_c}} = m\frac{\omega_c}{\omega}$

$d = \frac{\pi c}{\omega_c}$ $\lambda = \frac{2\pi c}{\omega}$

$v_m = c\sqrt{1 - \frac{m^2\omega_c^2}{\omega^2}} = c\sqrt{1 - \sin^2\theta_m} = c\sqrt{\cos^2\theta_m} = c\cos\theta_m$

$\theta_m = \arcsin\left(m\frac{\omega_c}{\omega}\right)$ $\cos\left(\arcsin\left(m\frac{\omega_c}{\omega}\right)\right)$

maior o m → maior arcsin → maior $\theta_m$ → menor o cos → menor $v_m$

velocidade maior para menor m

" menor para maior m

Para $m = 31 \rightarrow v = 0{,}58\times 10^8$ m/s

$\sin\theta_m = \frac{m\lambda_0}{n_1 d} = 0{,}98$

$\theta_m = 78{,}86°$

Para $m = 1 \rightarrow v = 2{,}998\times 10^8$ m/s

$\theta_m = 1{,}81°$

$\sin\theta_m = 0{,}03$

c) $\Delta t = t_{max} - t_{min} = \frac{1}{v_{min}} - \frac{1}{v_{max}} = \frac{1}{0{,}58\times 10^8} - \frac{1}{2{,}998\times 10^8} = 1{,}39\times 10^{-8} = 13{,}9$ ns

Formulário

6. Campo magnético → $\vec{H_x}$

$-i\omega\mu H = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \Leftrightarrow -i\omega\mu H_x = \frac{\partial(a_m U_m(y)\,e^{-i\beta z})}{\partial y} - \frac{\partial(a_m U_m(y)\,e^{-i\beta z})}{\partial z}\cot\theta \Leftrightarrow -i\omega\mu H_x = a_m e^{-i\beta z}\left(\frac{\partial U_m(y)}{\partial}\right) - a_m U_m(y)\cot\theta\,(-i\beta)\,e^{-i\beta z}$

$E_y = E\cos\theta_m$

$E_z = E\sin\theta_m$

$\frac{E_y}{E_z} = \frac{\cancel{E}\cos\theta_m}{\cancel{E}\sin\theta_m} \Rightarrow E_y = E_z\cot\theta$

$\Rightarrow H_x = -\frac{a_m}{i\mu\omega}e^{-i\beta z}\left[\frac{\partial U_m(y)}{\partial y} + i\beta U_m(y)\cot\theta\right]$

7. $S_z = \frac{1}{2}\mathrm{Re}(E_x H_y^*) = \frac{1}{2}\left(\sum_{m=0}^{M} a_m U_m(y)\cdot\frac{1}{\sqrt{\frac{\mu}{\varepsilon}}}\cos\theta_m\,a_m U_m(y)\right) = \frac{1}{2\eta}\cos\theta_m\sum_{m=0}^{M}|a_m|^2|U_m(y)|^2$

ver exercício 1

$P_z = \int_{-\infty}^{+\infty}\sum_{m=0}^{M}|a_m|^2\underbrace{|U_m(y)|^2}_{1}\frac{\cos\theta_m}{2\eta}\,dy = \sum_{m=0}^{M}|a_m|^2\frac{\cos\theta_m}{2\eta}$